

VISHWANATH VINOD

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EDUCATION

Indian Institute of Technology Madras

2022 - Present

Dual Degree (B.Tech in Electrical Engineering + M.Tech in Data Science & Artificial Intelligence)

9.74/10.0

Kuriakose Elias HSS Mannanam

2020 - 2022

Class XII - Kerala HSE

99.83%

PUBLICATIONS

Point Neuron Embedded Kalman Filter for Narrowband Sound Source Tracking

EUSIPCO 2026

Vishwanath Vinod, S. Xu, P. Samarasinghe, A. Bastine, Thushara Abhayapala

Cooperative Multi-Agent Reinforcement Learning for Fair Dynamic Pricing

Computing Conference 2026

Vishwanath Vinod, Rachel Kalpana Kalaimani

AWARDS

Future Research Talent Award

2025

Awarded by the Australian National University during my research internship

Research Spotlight Certificate of Excellence

2024 - 2025

Awarded for the best undergraduate research projects in the department

Sri V. Rajagopalan Memorial Award

2023 - 2024

Awarded for the highest academic performance in the department during sophomore year

RESEARCH EXPERIENCE

Privatizing State Space Models

Mar '26 - Present

Masters Project Under Prof. Krishna Pillutla

- Conducted an extensive literature review on State Space Models (SSMs) and delivered a series of **research talks** on the topic.
- Implemented **MAMBA from scratch** in PyTorch, including the parallelized selective scan algorithm.
- Incorporated **Privacy amplification by iteration** into the contractive steps of SSMs and derived privacy guarantees for a single-layer.
- Characterized privacy leakage and extended it to **multi-layer SSMs**, including models with input-dependent parameters like MAMBA.
- Empirically validated the synthetic text generation performance of this private model by evaluating it on the **WikiText datasets**.

Point Neuron Embedded Kalman Filter (PNEKF)

May '25 - Sep '25

Research Intern, Australian National University under Prof. Thushara Abhayapala

- Implemented Point Neuron Learning in PyTorch and extended it to **3-D sound field reconstruction** in reverberant settings.
- Designed a Point Neuron-based **Orthogonal Matching Pursuit (OMP)** algorithm for sound source localization to initiate tracking.
- Proposed a **novel Kalman Filter variant** that integrates the point neuron framework to facilitate **multi-source tracking**.
- Simulated reverberant room acoustics using the **Image Source Model (ISM)** pipeline in MATLAB for empirical evaluation.
- **Experimentally validated** PNEKF's superiority over baseline methods across diverse frequencies, sources, and microphone settings.

Cooperative Multi-Agent Reinforcement Learning for Fair Dynamic Pricing

Apr '24 - Nov '24

Undergraduate Research Project under Prof. Rachel Kalpana Kalaimani

- Designed a custom OpenAI Gym environment for dynamic pricing with **finite inventory** and a **limited selling horizon**.
- Conducted a comprehensive literature review on dynamic pricing, fairness metrics, and MARL methods (**IRAT**, **Dec-MADDPG**).
- Developed a **cooperative MARL** framework with a **dual-reward** formulation to promote **group fairness** in dynamic pricing.
- Introduced a novel **control parameter** to tune the fairness-profit trade-off in this multi-objective optimization setting.
- **Experimentally validated** the proposed algorithm's superiority through a comprehensive analysis against the existing benchmark.

RELEVANT COURSEWORK

MACHINE LEARNING

Introduction to Reinforcement Learning, Privacy in AI, Advanced Optimization, Natural Language Processing
Learning from Human Preferences, Deep Learning for Imaging, Introduction to Machine Learning,
Applied Programming Lab, Deep Learning Specialization, Data Analytics Laboratory

ELECTRICAL ENGINEERING

Control Systems & Engineering, Signals & Systems, Digital Signal Processing, Image Signal Processing
Communication Systems, Analog Systems & Lab, Electrical Machines & Lab, Microprocessor & Lab

MATHEMATICS

Applied Linear Algebra for Electrical Engineering, Probability Foundations for Electrical Engineers,
Statistics for Electrical Engineers, Computational Physics, Differential Equations

COMPUTER SCIENCE

Object Oriented Programming Specialization, Data Structures & Algorithms, Computer Organization

TEACHING ASSISTANT

DA6400: Introduction to Reinforcement Learning (Prof. Balaraman Ravindran)

Jan '26 - May '26

DA2401: Foundations of Machine Learning (Prof. Krishna Pillutla)

Jul '26 - Present

SCHOLASTIC ACHIEVEMENTS

OPENING RANK	Enrolled in the Electrical Engineering department at IIT Madras as the department's highest-ranked admit.
JEE ADVANCED 2022	Secured All India Rank 252 among 200,000 candidates, placing in the top 0.2 %tile of candidates
JEE MAIN 2022	Secured All India Rank 506 among 1.1 million candidates, with a percentile score of 99.95 %tile
KVPY 2020	Secured All India Rank 30 (SA stream); received the KVPY Fellowship from the Government of India
KEAM 2022	Secured State Rank 1 in the Kerala Engineering Entrance Exam among 73,000 candidates
OLYMPIADS	Qualified for the Indian National Mathematics Olympiad 2019 and the National Junior Science Olympiad 2018 .

MAJOR COURSE PROJECTS

Reinforcement Learning Methods

Jan '25 – May '25

DA6400 - Introduction to Reinforcement Learning under Prof. Balaraman Ravindran

- Conducted experimental studies of multiple RL algorithms across diverse use cases.
- Analyzed the performance of **SARSA** and **Q-Learning** on a modified grid-world under varied configurations.
- Implemented **Dueling-DQN**, **MC-Reinforce**, and **Actor-Critic** methods on OpenAI Gymnasium tasks (*CartPole*, *AcroBot*, *MountainCar*).
- Performed comparative evaluation of hierarchical RL algorithms such as **SMDP** and **Intra-option Q-Learning** on the Taxi environment.

Information Retrieval System

Jan '25 – May '25

CS6370 - Natural Language Processing under Prof. Sutanu Chakraborty

- Implemented and evaluated classical (**TF-IDF**, **BM25**), semantic (**LSA**, **ESA**, **NESA**), and hybrid IR models on the Cranfield dataset.
- Incorporated semantic enhancements such as **WordNet-based query expansion** and dense transformer embeddings (**All-MiniLM**).
- Designed a modular preprocessing pipeline with sentence segmentation, tokenization, stemming, and stopword removal.
- Built **query autocomplete** using N-gram and Trie models with a top-*k* interactive interface robust to typos and incomplete input.
- Conducted comparative evaluation with **MAP@10**, **Precision@10**, **Recall@10**, **nDCG@10**, and significance tests on query-level scores.

Image Denoising & Privacy in Lensless Imaging

Jan '25 - May '25

EE5175 - Image Signal Processing & EE5179 - Deep Learning from Imaging under Prof. A.N. Rajagopalan

- Implemented **BM3D** to denoise images from the **RENOIR dataset** and low-light smartphone video frames.
- Benchmarked BM3D against classical filters (NLM, Bilateral, Gaussian) using **PSNR** and **SSIM** metrics.
- Implemented classical image processing tasks including segmentation, mosaicing, occlusion detection, deblurring and depth mapping.
- Surveyed lensless imaging architectures (**FlatCam**, **PhlatCam**, **FlatNet**, **OpEnCam**, **CAnOpIC**) and face recognition models.
- Conducted ablation studies of these models (**DeepFace**, **ArcFace**) on LFW dataset under varying noise levels in **Wiener deconvolution**.

Machine Learning Techniques

Jan '25 - May '25

DA5401 - Data Analytics Lab under Prof. Sudarshan Santhiappan

- Mitigated class imbalance via **Clustering (CBO, CBU)**, **SMOTE**, and **GMM-based synthetic resampling**.
- Analyzed multi-label manifold structures and data veracity using **t-SNE** and **Isomap** embeddings.
- Implemented data imputation strategies including **Linear/Non-Linear Regression**, Median, and Listwise deletion.
- Evaluated multi-class model performance using macro/micro **ROC** and **Precision-Recall** curves.
- Built **similarity regression pipelines** for AI prompt-responses under extreme class imbalance and limited sample constraints.
- Boosted accuracy using rank normalization, metric shuffling, dimensionality reduction, and ensemble modeling techniques.

TALKS & PRESENTATIONS

2025	Research Presentation @ Audio & Acoustic Signal Processing Group, Australian National University
2026	Speaker at the Electrical Engineering Research Symposium @ IIT Madras
2026	Oral Presentation at the 14th Computing Conference @ London, UK
2026	State Space Models Research Talk @ Machine Learning & AI Lab, IIT Madras

Co-CURRICULARS

2024–25	Coordinator @ Electrical Engineering Research Club, IIT Madras
2023–24	Deputy Placement Coordinator @ Placement & Internship Cell, IIT Madras
2022–24	Coordinator @ Coding & Logic, Shaastra Tech Fest 2024, IIT Madras